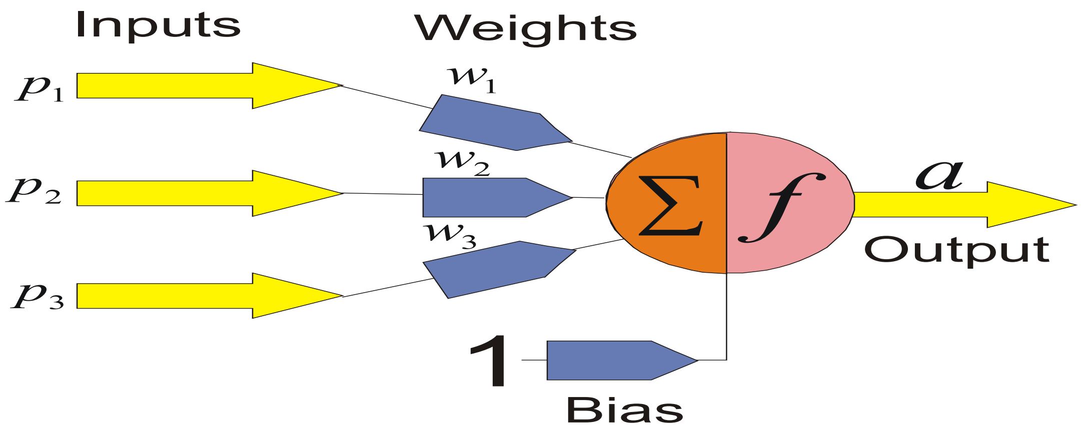
The Key Elements of Neural Networks

- Neural computing requires a number of neurons, to be connected together into a neural network. Neurons are arranged in layers.

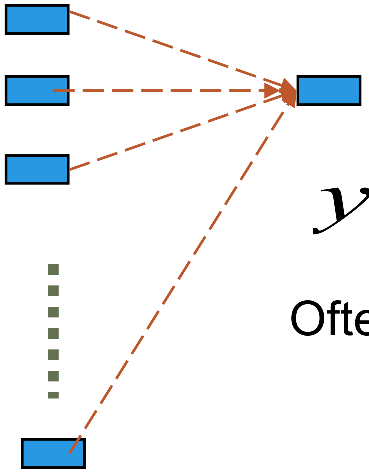


$$a = f(p_1 w_1 + p_2 w_2 + p_3 w_3 + b) = f\left(\sum p_i w_i + b\right)$$

- Each neuron within the network is usually a simple processing unit which takes one or more inputs and produces an output. At each neuron, every input has an associated weight which modifies the strength of each input. The neuron simply adds together all the inputs and calculates an output to be passed on.

Properties of Neural Networks

- No connections within a layer
- No direct connections between input and output layers
- Fully connected between layers
- Often more than 3 layers
- The number of output units need not be equal to the number of input units
- Number of hidden units per layer can be more or less than input or output units

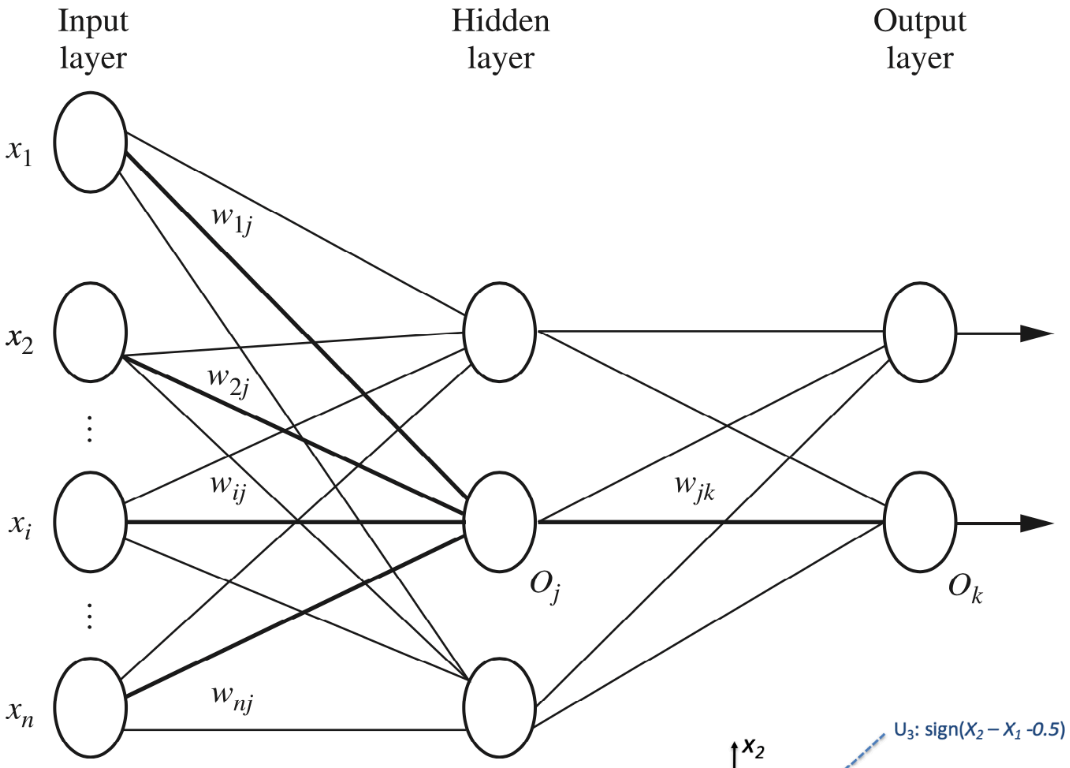


Each unit is a perceptron

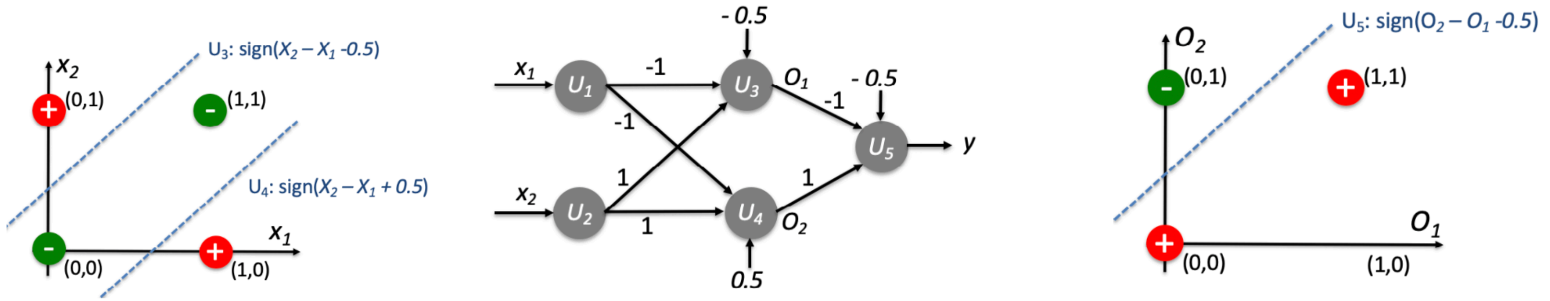
$$y_i = f\left(\sum_{j=1}^{m} w_{ij} x_j + b_i\right)$$

Often include bias as an extra weight

Feed-forward Neural Networks

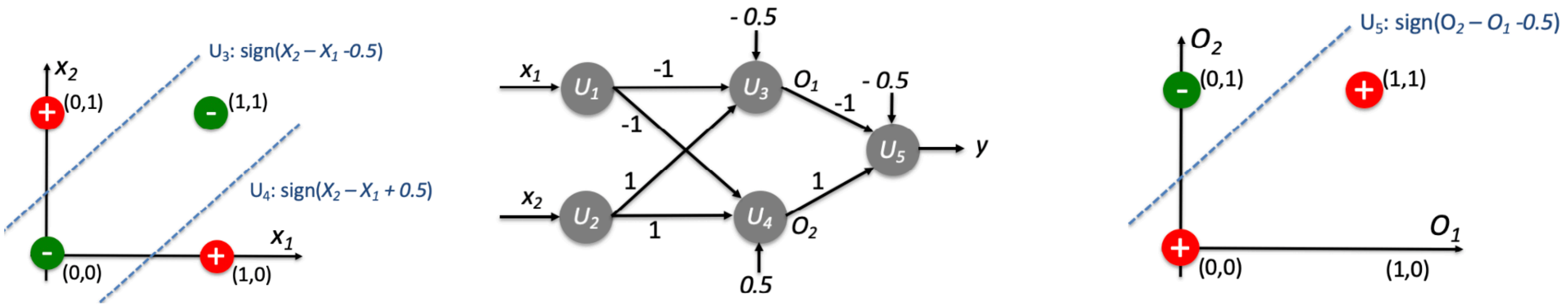


- Why multiple layers
  - Automatic feature/representation learning
  - Learn complicate (nonlinear) mapping function



(a) Input 4 tuples (b) MLP (c) Outputs of two hidden units

Feed-forward Neural Networks (cont.)

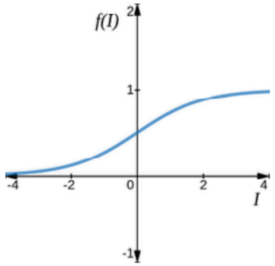
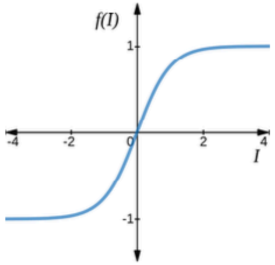


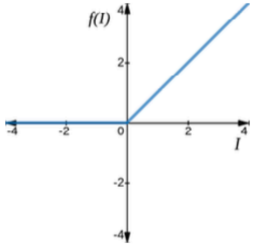
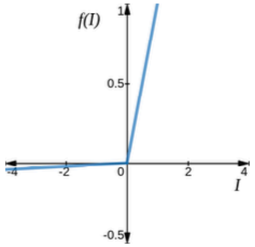
(a) Input 4 tuples (b) MLP (c) Outputs of two hidden units

X_1	X_2	O_1 = sign ($X_2 - X_1$-0.5)	O_2 = sign ($X_2 - X_1$ + 0.5)
0	0	0	+1
1	1	0	+1
1	0	0	0
0	1	1	1

Typical Non-linear Activation Functions

- Examples of activation functions

Sigmoid	$\frac{1}{1+e^{-I}}$	
Tanh	$\frac{e^{I}-e^{-I}}{e^{I}+e^{-I}}$	

ReLU	$\begin{cases} 0, I \leq 0 \\ I, I > 0 \end{cases}$	
Leaky ReLU	$\begin{cases} 0.01 \times I, I < 0 \\ I, I \geq 0 \end{cases}$	
ELU	$\begin{cases} \alpha(e^{I} - 1), I \leq 0 \\ I, I > 0 \end{cases}$